Bio-Signal Studies for a Relaxation-Based Biofeedback Training of Robot Users in Surgical Tasks

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Abstract— **Robot-assisted surgery enhances accuracy and safety, yet operator stress and attention remain critical determinants to human–robot system performance. Biofeedback, the real-time presentation of physiological signals to facilitate self-regulation has been proposed as a potentially effective training strategy. This preliminary study integrated heart rate variability (HRV) and electroencephalography (EEG) into a robot-user training platform, in which deviations from calibrated thresholds triggered scalpel retraction. Twenty-four naive participants trained under three conditions (no feedback, HRV feedback, EEG feedback) and then performed standardized test tasks measuring accuracy and and consistency.**

The results suggested that EEG feedback yielded higher accuracy and smoother execution, while HRV feedback offered complementary information on stress regulation. Subjective reports indicated reductions in perceived workload and improvements in self-awareness, though some participants found the interruptions disruptive. An alternative analysis confirmed that HRV and EEG capture distinct dimensions of the psychophysiological state, suggesting benefits of combining them in multi-signal biofeedback.

Overall, the findings provide initial evidence for the feasibility of biofeedback for robot-user training and highlight its potential for future co-designed, adaptive systems supporting surgical education and practice.

I. INTRODUCTION

A. *Background*

The field of robot-assisted surgery has advanced significantly over the past few decades. By integrating robotic systems into clinical practice, modern medicine has achieved unprecedented accuracy and consistency in surgical procedures, enabling minimally invasive surgery (MIS) [1][2] and improved patient outcomes [3]. Platforms such as the da Vinci surgical robot improve dexterity and reduce tremor [4], yet system performance ultimately depends on the cognitive and physiological state of the human operator. Stress, fatigue, and cognitive overload can impair motor coordination, decision-making, and attentional focus, posing risks to procedural success and patient safety [5].

Research across high-stakes environments, including neurosurgery [6] and operator stress modelling [7], shows that electroencephalography (EEG) and heart rate variability (HRV) can effectively monitor stress and attention, with elevated stress correlating with increased error rates. Physiological signals are thus not merely passive indicators but can be harnessed to support real-time adaptation and training. Biofeedback, broadly defined as the real-time presentation of physiological signals to facilitate self-regulation, has been successfully applied in domains such as cognitive rehabilitation [8]. However, most surgical training platforms continue to prioritize technical skill acquisition while overlooking the dynamic interplay between operator state and system performance.

Evidence further indicates that stress-induced physiological changes degrade fine motor control and visuospatial reasoning, both essential in robot-assisted surgery. Loeb et al. reported that surgeons under time pressure exhibited diminished psychomotor performance, linked to fluctuations in autonomic nervous system activity [9]. Similarly, Dias et al. found that heightened sympathetic arousal during laparoscopic procedures correlated with longer task times and reduced accuracy [10], reinforcing the value of monitoring stress-related markers during training.

EEG-derived indices are associated with attentional engagement and working memory load [11], while HRV reflects autonomic balance and mental workload [12]. When analyze together, EEG and HRV provide a more comprehensive representation of a trainee's cognitive-affective state, enabling richer opportunities for adaptive training. Applications in aviation and astronaut training already demonstrate the utility of biofeedback for enhancing performance and regulating stress [13][14]. Yet, integration into surgical training remains limited, with simulators rarely incorporating physiological or cognitive-emotional feedback loops [15][16]. This gap is significant, as many surgical errors stem not from technical deficits but from lapses in attention or stress mismanagement [17]. Addressing it requires training systems that integrate physiology-aware feedback to strengthen both technical performance and resilience under pressure.

The potential applications of biofeedback in other high-stakes domains further demonstrates its value. In aviation and astronaut training, biofeedback has been employed to improve performance and support stress regulation [13][14]. Despite this evidence, integration of biofeedback into surgical training remains limited. Contemporary simulators emphasize procedural steps, haptic interaction, and visual guidance, but rarely incorporate feedback loops based on physiological or cognitive-emotional states [15][16]. This gap is particularly concerning given that many surgical errors arise not from deficits in motor skill but from lapses in attention or inadequate stress management [17]. Addressing this limitation requires the development of training systems that integrate physiology-aware feedback to strengthen both technical performance and resilience under stress.

B. *Motivation*

Optimal performance in robot-assisted surgery requires not only technical competence but also sustained cognitive clarity, attentional control, and physiological stability in

high-pressure environments. Traditional training paradigms often neglect these dimensions, overlooking the cognitive and affective demands of clinical practice [18]. To address this limitation, the present study introduces a surgical robot-user training system integrating real-time HRV and EEG biofeedback.

Unlike prior research that mainly correlated physiological signals with performance metrics, this system implements closed-loop biofeedback: deviations from predefined stress or attentional thresholds trigger adaptive modifications, specifically a temporary scalpel retraction. This pause encourages trainees to regain composure before resuming, discouraging operation under suboptimal conditions. By coupling real-time feedback with adaptive scenarios, the system fosters self-regulation strategies essential for sustaining accuracy and resilience. In aligning technical skill acquisition with physiological and cognitive awareness, the approach advances a more comprehensive framework for surgical education.

C. *Aims*

1) To investigate how operator stress and attention levels influence accuracy in robotic surgical tasks (measured via EEG and HRV).
2) To evaluate the effectiveness of real-time biofeedback in promoting physiological regulation and improving human–robot system performance.
3) To assess user adaptation strategies in response to biofeedback-triggered task interruptions.

D. *Objectives*

1) Design and implement a surgical robot-user training platform integrating EEG and HRV sensors with a Phantom Omni haptic device in Unity.
2) Develop a biofeedback control system that modulates scalpel behaviour based on real-time physiological thresholds.
3) Evaluate biofeedback impact on surgical task performance under three conditions (no feedback, HRV, EEG).
4) Analyze objective performance metrics, physiological trends, and subjective user feedback to evaluate biofeedback efficacy.
5) Perform statistical analyses to correlate physiological states with task outcomes and derive implications for adaptive surgical training systems.

In summary, integrating bio-signals into a surgical robot–user training platform constitutes a significant contribution to human–robot interaction (HRI). The proposed approach supports the development of next-generation simulators that are not only technically robust but also physiologically adaptive, responding in real time to fluctuations in the trainee's cognitive and affective state. By linking technical skill acquisition with psychophysiological self-regulation, such systems can enhance learning outcomes and operational safety, fostering greater resilience in surgical performance.

II. RESEARCH METHODOLOGY

A. *Implementation*

1) *System Architecture*

The system integrates three primary components:

- A non-immersive virtual surgical training environment built in Unity 3D and C# scripting, incorporating bio-signal controls for adaptive feedback.
- A haptic input device, Touch (also name as Phantom Omni, by 3D system) for user control.
- A multimodal bio-signal processing architecture: The system acquires physiological signals through a Polar H10 chest strap (for HRV) and an Emotiv MN8 earphone (for EEG). The raw data are processed in real time using Python algorithm pipelines, which extract key biomarkers: HRV features (RMSSD) of cardiac activity and EEG band-power to estimate cognitive attention states. Processed signals are transmitted via a custom UDP protocol for data streaming, enabling seamless integration with the training environment.

This architecture ensures reliable real-time integration of physiological data with the surgical simulation environment while maintaining the accuracy consistency required for closed-loop biofeedback applications.

2) *HRV Measurement - RMSSD*

Heart Rate Variability (HRV) is a widely used physiological measure for assessing autonomic nervous system (ANS) activity and has been shown to reflect both emotional and cognitive states [19]. In the present study, the Root Mean Square of Successive Differences (RMSSD) is selected as the primary HRV feature due to its high sensitivity to parasympathetic (vagal) activity, which is closely associated with relaxation and recovery processes [20]. By focusing on RMSSD, the study aims to capture subtle variations in autonomic regulation that may indicate stress or attentional fluctuations during surgical training.

The HRV signal was acquired using a Polar H10 chest strap [21], which records RR intervals and streams data via Bluetooth. Although not a clinical-grade ECG, it is a validated, low-cost, and wearable device widely used in sports science and HCI research. Data parsing and feature extraction were implemented in Python, using open-source libraries.

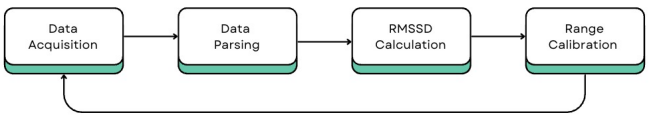


Fig. 1. HRV data processing flow chart.

Fig. 1 is the flow chart of HRV data processing, which contains 4 steps:

- Data Acquisition: RR intervals were captured from the Polar H10 chest strap via Bluetooth, representing the time difference between successive R-waves in the ECG signal.

- Data Parsing: The byte stream is decoded to extract individual RR intervals in milliseconds.
- RMSSD Calculation: Calculate Root Mean Square of Successive Differences (RMSSD) by using a sliding window.

$$RMSSD = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1}(RR_{i+1} - RR_i)^2}$$

- Range Calibration: Participant-specific thresholds were established during a 5-minute guided breathing session (6 breaths/min). Mean and standard deviation were calculated, and adaptive thresholds were defined as the equation, with $k = 1.0$.

$$\text{Threshold}_{\text{low}} = \mu_{\text{RMSSD}} - k \cdot \sigma_{\text{RMSSD}}$$

$$\text{Threshold}_{\text{high}} = \mu_{\text{RMSSD}} + k \cdot \sigma_{\text{RMSSD}}$$

*3) **EEG Measurement - Attention***

Electroencephalography (EEG) is a widely used neurophysiological measure that reflects the electrical activity of the brain, originating from synchronized postsynaptic potentials of cortical neurons [22]. EEG has long been employed to investigate cognitive workload, emotional regulation, and mental fatigue, making it particularly suitable for applications in human–machine interaction [23]. In the present study, an attention index derived from EEG signals is selected as the primary marker for detecting human cognitive control [24].

EEG signals were acquired using the Emotiv MN8 earphone device, which provides proprietary metrics (attention) through the Emotiv Cortex API [25]. Although the internal algorithms are not disclosed, the device was chosen because it is low-cost, wearable, and accessible, offering practical advantages for preliminary studies on physiology-aware training. Real-time communication with the Emotiv Cortex API was implemented via a WebSocket interface, enabling continuous acquisition and processing of attention values. EEG data processing is similar to the HRV process, including three steps: data acquisition, data handling, and range calibration.

*4) **Feedback Mechanism - Unity 3D animation***

The scalpel disappearance animation functioned as the core biofeedback mechanism within the training environment. When a participant's HRV or EEG signals fell outside their calibrated thresholds, the virtual scalpel faded from view, temporarily preventing further task execution. Once the participant's physiological state returned within range, the scalpel reappeared, enabling continuation of the task. This dynamic, closed-loop mechanism directly linked physiological regulation to task performance, thereby reinforcing self-regulation and fostering greater attentional and emotional awareness during training.

B. Experiment Methodology

*1) **Conditions***

The experiment consisted of two distinct biofeedback conditions designed to evaluate the effect of physiological regulation on task performance:

- Condition A – HRV Biofeedback: When participants' HRV RMSSD value fell outside their personalized calibration range, the virtual scalpel would automatically disappear, providing real-time feedback on physiological state.
- Condition B – EEG Biofeedback: When participants' EEG-derived attention value dropped below the personalized lower calibration threshold, the virtual scalpel would disappear, providing real-time feedback on cognitive engagement.

*2) **Procedure***

Each participant followed a fixed protocol:

- Range Calibration: A 5-minute calibration session was conducted to establish baseline measurements for HRV and EEG-derived attention indices. Participants were instructed to maintain a relaxed state while following paced breathing at 6 breaths per minute to standardize autonomic nervous system influences.
- Training Period: Each participant completed two training blocks, one under each biofeedback condition. Within each block, they performed 10 training trials under the assigned condition, followed by 3 testing trials without biofeedback. This design allowed performance under feedback to be compared directly with baseline conditions.
- Counterbalancing: To minimize practice effects and proficiency bias, participants were randomly assigned to one of two condition sequences: A→B or B→A.
- Post-Training Assessment: After each block, participants completed standardized questionnaires evaluating workload, usability, and subjective biofeedback experiences.

*3) **Task Design***

The virtual environment was designed to replicate key aspects of surgical instrument control while remaining accessible for novice participants. Using the Phantom Omni (Touch) haptic device, participants manipulated a virtual scalpel with force-feedback. To execute each movement, the device's input button had to be continuously depressed while tracing a predefined trajectory, simulating the accuracy and control required in robotic-assisted surgical tasks.

The study employed three distinct trajectory types, shown in Fig. 2:

- Type A: An irregular cloud-shaped contour combining C-shaped and S-shaped compound curves.
- Type B: A concentric circular pattern with multiple nested rings, requiring continuous spiral motion.
- Type C: A straight line, introduced only in testing as a novel trajectory.



Fig. 2. Three type of trajectories.

Fig. 3 is an example of the experiment. During the training phase, participants practices two trajectories (Types A & B), completing five trials of each, with a maximum duration of 50 seconds per trial. In the testing phase, they performed one trial for each trained trajectory plus an additional trial with the novel trajectory (Type C). This design allowed for direct performance comparisons between trained conditions (A vs. B) and assessment of skill transfer to untrained movements (A/B vs. C). To minimize practice effects, task presentation order was counterbalanced across participants.



Fig. 3. Example of actual user operation.

4) *Assessment Metrics*

The study incorporated multimodal evaluation through both objective performance measures and subjective experience reports:

1) Objective Measures
   a) Trajectory Mean Absolute Error (MAE): Millimetre-level deviation from ideal path.
   b) Retraction Frequency: Count of forced scalpel disappearances triggered by biofeedback threshold violations.
   c) Completion Time: Total trajectory duration.
   d) HRV & EEG Parameters: RMSSD and attention value during task execution.
2) Subjective Measures: Post-task questionnaires, including SUS and NASA-TLX scale to assess participants' stress, difficulty, and engagement.

5) *Data Processing*

The raw EEG and HRV data were stored in CSV (time-series) formats and processed using Python for preprocessing, normalization, and statistical analysis. Outliers beyond ±3.0 standard deviations were excluded. Performance scores were normalized to allow between-subject comparisons.

III. RESULTS

A. *Participants*

A total of 24 participants (12 male, 12 female) were recruited, with an average age of 24.5 years (SD = 2.7, range 22–33). All participants were healthy adults with no reported neurological or cardiovascular disorders. Importantly, participants were naive with respect to surgical training and robotic systems, ensuring that performance outcomes reflected baseline learning rather than prior expertise. This recruitment strategy aligns with the preliminary nature of the study, providing a controlled first step before extending to surgical trainees or professionals in future work.

B. *Biofeedback Calibration*

1) *HRV - RMSSD Calibration*

The RMSSD-based HRV calibration successfully established individualized low and high thresholds for all participants. Across participants, The HRV calibration thresholds ranged from an average low of 14.16 ms (SD = 9.63) to a high of 42.07 ms (SD = 19.58). The left diagram in Fig.4 shows the distribution of participant HRV thresholds.

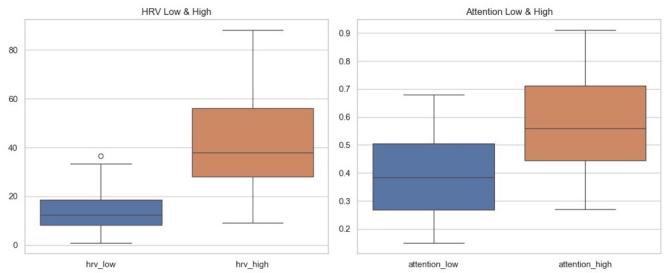


Fig. 4. Distribution of HRV and EEG Range Calibration.

2) *EEG - Cognitive Attention Calibration*

EEG attention thresholds were similarly individualized. Across participants, the mean attention low threshold was 0.396 (SD = 0.16) and the mean attention high threshold was 0.579 (SD = 0.18). The right diagram in Fig.4 shows the distribution of participant HRV thresholds.

C. *Task Performance Metrics*

To evaluate participant performance under biofeedback control, multiple objective metrics were collected across both training and test phases. Including trajectory MAE, task completion time and scalpel retraction count.

In this study, to quantify the similarity between participants' drawn trajectories and the reference path, computed the symmetric difference area. Specifically, both the participant's trajectory and the reference trajectory were converted into buffered regions, and the non-overlapping area between these two regions was measured. This metric captures the geometric discrepancy between the trajectories by considering not only point-wise deviations but also shape differences, such as overshooting or wobbling along the path [26]. In the following content, all 'MAE' refer to symmetric difference area.

1) *Training Performance*

Fig. 5 and Fig. 6 compare the performance of two trajectory types, Type A (Trials 1-5) and Type B (Trials 6-10), focusing on MAE and completion time.

Through observation, find that for Type A, the Area Difference starts higher at about 4.0 mm^2 and gradually

improves to approximately 3.2 mm^2 by Trial 5, demonstrating enhanced accuracy with practice. In contrast, Type B maintains consistently better accuracy, with Area Difference values ranging between 3.0 to 3.2 mm^2 throughout all trials.

For completion time, Type A shows improvement from about 36 seconds down to 32 seconds across trials, indicating increasing efficiency. Type B demonstrates stable performance with completion times between 31-32 seconds, which is slightly but consistently faster than Type A's best performance.

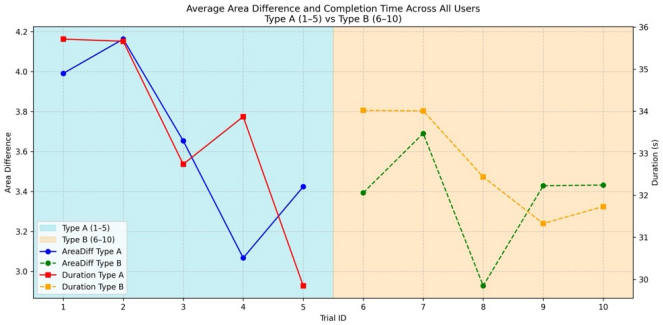


Fig. 5. Average MAE and Completion Time Trend Across All Users.

Participants showed progressive reductions in MAE and completion time across the 5 training trials, indicating motor learning and adaptation. Type B outperforms Type A in both accuracy (lower MAE) and speed (shorter duration). The gap suggests fundamental differences in task complexity or user familiarity.

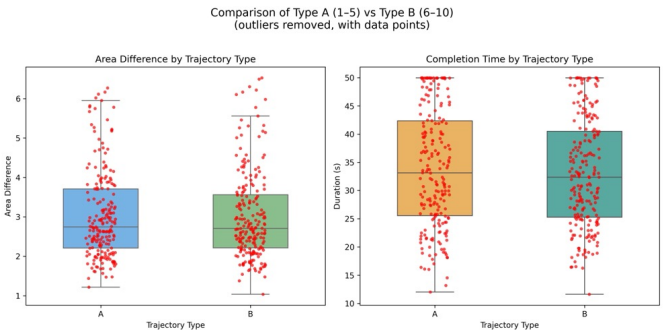


Fig. 6. Average MAE and Completion Time Trend Across All Users.

Table I compares the performance of Trajectory Types A and B. Type B shows slightly lower MAE and Completion Time compared to Type A, with p-values of 0.8861 and 0.3514 respectively, indicating there is a statistically significant differences between the two types.

TABLE I
PERFORMANCE COMPARISON BETWEEN TRAJECTORY TYPES A AND B

Metric	Type A	Type B	p-value
MAE (mm^2)	3.03 ± 1.12	3.01 ± 1.11	0.8861
Completion Time (s)	33.87 ± 10.42	32.97 ± 9.64	0.3514

2) Compare Training and Testing Performance

Participants performed a series of training trials on two distinct surgical trajectories (Type A & B), completing five practice trials and one test trails per trajectory. The mean and standard deviation of key metrics during training and testing are summarized in Table II.

According to the result, find that:

- Trajectory Accuracy: MAE decreased from training to testing for both trained trajectories (Type A & B), indicating improved accuracy. Type C, although untrained, had a lower MAE, suggesting participants transferred learned skills effectively.
- Completion Time: Both trained trajectories showed reductions in completion time, while Type C trials were completed much faster, reflecting ease of execution or simplified task structure.

TABLE II
PERFORMANCE METRICS ACROSS TRAJECTORY TYPES

Trajectory Type	Training Phase	Test Phase
Trajectory MAE (mm)		
Type A	3.66 ± 2.00	3.14 ± 1.28
Type B	3.38 ± 1.74	2.77 ± 1.10
Type C	–	1.08 ± 0.58
Completion Time (s)		
Type A	33.57 ± 12.11	30.04 ± 7.24
Type B	32.71 ± 10.47	32.05 ± 9.06
Type C	–	12.01 ± 6.44

3) HRV & EEG Trends

Biofeedback signals were monitored during both training and testing phases to evaluate physiological self-regulation under different trajectory conditions. Average HRV RMSSD and EEG attention values are reported in TABLE III.

TABLE III
PERFORMANCE METRICS ACROSS DIFFERENT TYPES

Type	Retraction Count	RMSSD (ms)	Attention
A Train	0.89 ± 1.17	28.64 ± 13.35	0.60 ± 0.16
A Test	/	29.30 ± 13.86	0.57 ± 0.14
B Train	0.83 ± 1.06	28.73 ± 14.11	0.60 ± 0.14
B Test	/	27.47 ± 11.70	0.57 ± 0.15
C Test	/	29.48 ± 15.93	0.55 ± 0.14

Through observation, find that:

- HRV Trends: RMSSD values remained relatively stable across trajectories and phases, suggesting participants maintained similar parasympathetic activity during performance. Slight increases were observed in Type C trials during testing, potentially reflecting lower cognitive load due to simpler trajectory or transfer learning effects.
- EEG Attention Trends: Attention values decreased slightly from training to testing for trained trajectories, which may indicate habituation or reduced cognitive effort once tasks became familiar. Type C testing trials

also showed slightly lower attention, consistent with a less demanding task.

- Scalpel Retractions: Retraction counts remained low overall (<1.1), indicating effective physiological self-regulation and stable biofeedback control.

Fig. 7 is the overall results of different bio-feedback control condition. Compared to HRV, EEG demonstrates superior performance across all measured metrics, achieving a 4.2% improvement in accuracy and a 9.0% reduction in procedure time. Most significantly, Condition EEG substantially reduced scalpel retractions by 65.1%, lowering them from 0.86 to 0.3 instances. These results collectively indicate that Condition EEG offers enhanced surgical accuracy, greater operational efficiency, and markedly improved procedural consistency

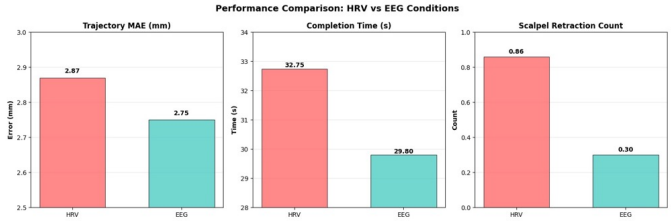


Fig. 7. HRV & EEG Performance Comparision.

D. Correlation Analysis

The Pearson correlation coefficient (r) is a statistical measure that describes the strength and direction of the linear relationship between two continuous variables [27]. In TABLE IV, it is used to examine relationships between physiological signals and task performance metrics, where $r > 0$ indicates positive correlation, and $r < 0$ indicates negative correlation.

- HRV RMSSD vs. Trajectory MAE: Weak negative correlation (r = -0.12), suggesting higher parasympathetic activity may slightly relate to improved accuracy.
- EEG Attention vs. Trajectory MAE: Moderate negative correlation ($r = -0.34$), indicating that higher attention levels are associated with lower MAE and better surgical accuracy.
- HRV RMSSD vs. Completion Time: Near zero correlation ($r = 0.05$), suggesting completion speed was largely independent of HRV.
- EEG Attention vs. Completion Time: Weak negative correlation ($r = -0.18$), implying that participants with higher attention tended to complete tasks slightly faster.
- Scalpel Retraction Count vs. HRV/EEG: Low correlations ($|r| < 0.15$), indicating that retraction events were mostly influenced by trajectory difficulty rather than physiological signals.

The results indicate a moderate positive correlation (r = 0.42) between HRV and EEG across most conditions, suggesting that higher HRV values (indicative of relaxation) tend to align with higher EEG attention levels. While HRV exhibited more stability in detecting stress fluctuations,

TABLE IV
PEARSON CORRELATION COEFFICIENTS BETWEEN PHYSIOLOGICAL METRICS AND PERFORMANCE OUTCOMES

Metric Pair	r	Interpretation
HRV - MAE	-0.12	Weak negative
EEG - MAE	-0.34	Moderate negative
HRV - Time	+0.05	Very weak positive
EEG - Time	-0.18	Weak negative
HRV - Retraction	-0.08	Weak negative
EEG - Retraction	-0.14	Weak negative

EEG provided finer-grained resolution of attentional changes. These findings imply that EEG may serve not merely as a substitute but as a complementary measure to HRV, enabling a more robust multimodal biofeedback framework.

E. Alternative Analysis: HRV vs EEG

To further examine the relationship between EEG attention values and HRV (RMSSD), two complementary analyses were conducted: a Bland–Altman agreement test and a scatter-based correlation analysis.

Fig. 8 presents the Bland–Altman plot between standardized EEG and HRV values. The mean difference was effectively zero, indicating no systematic bias between the two measures at the group level. Most data points fell within the ± 1.96 SD limits (-2.79 to +2.79), suggesting an overall acceptable agreement. However, the wide spread of differences reveals notable variability across individuals, indicating that EEG and HRV signals do not consistently align at the single-subject level. This implies that, although both metrics reflect aspects of stress and attention, their agreement is limited in practical applications.

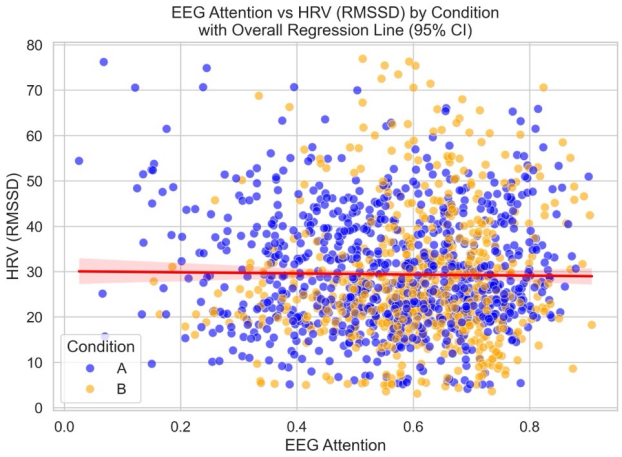


Fig. 8. Scatter Plot with Regression Line Comparing HRV and EEG.

Fig. 9 illustrates the scatter plot between EEG attention values and HRV RMSSD across both feedback conditions. The regression line (95% CI) is nearly flat, with a slope close to zero, suggesting that there is no significant linear relationship between EEG and HRV. Data points were

widely dispersed across conditions, further reinforcing that the two signals capture distinct physiological dimensions. Specifically, EEG appears to primarily track attentional focus, whereas HRV is more sensitive to autonomic stress responses.

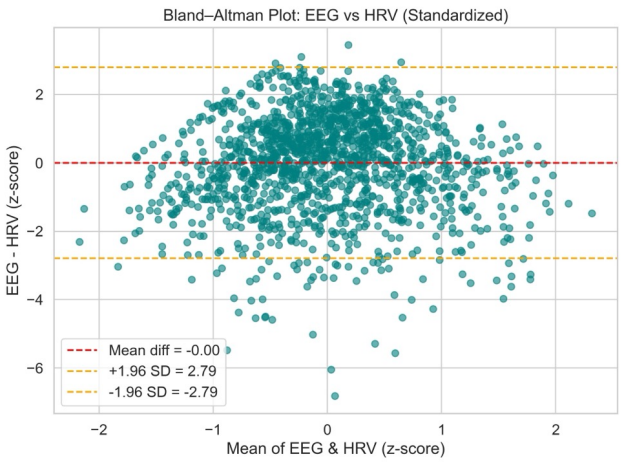


Fig. 9. Bland–Altman Plot of HRV and EEG.

Taken together, these analyses suggest that EEG cannot be treated as a simple replacement for HRV. While the two measures do not strongly correlate, their complementary nature indicates potential for multi-signal fusion. Integrating both EEG and HRV into composite indices could yield more reliable and robust biofeedback thresholds, reducing false positives and improving training efficacy in surgical simulation systems.

In summary, rather than considering EEG as a substitute for HRV, future research should emphasise their complementary value and investigate fusion-based approaches that leverage both signals for more adaptive and clinically relevant biofeedback designs.

F. Subjective Results

To complement the objective task performance metrics, participants completed post-experiment questionnaires evaluating platform usability, perceived workload, biofeedback and animation experience, and training effectiveness. TABLE V is the results.

1) System Usability (SUS)

Participants rated the system as generally intuitive and accessible. The mean rating for ease of use was high (M = 3.9 ± 0.8), while ratings for complexity were low (M = 2.1 ± 0.7), indicating that most participants did not perceive the platform as overly complicated. Confidence in operating the platform was strong (M = 4.2 ± 0.6), suggesting the interface and interaction design were effective for first-time users.

2) NASA-TLX

Workload assessment revealed a moderate cognitive and physical demand. Mental demand scored the highest (M = 6.5 ± 1.2), reflecting the attentional and motor-control requirements of surgical tasks. Physical demand and effort were moderate (M = 5.4 ± 1.5 and 5.8 ± 1.3, respectively), while temporal demand was lower (M = 4.2 ± 1.8), suggesting that participants did not feel excessively rushed. Importantly, self-rated performance was high (M = 7.6 ± 1.1), and frustration remained low (M = 2.9 ± 1.4), indicating positive user experience under both biofeedback conditions.

3) Platform Experience: Biofeedback & Animation

Most participants reported that biofeedback increased awareness of their physiological states (M = 3.8 ± 0.9) and helped them maintain focus and relaxation (M = 3.6 ± 1.0). Distraction levels were low (M = 2.1 ± 0.7), suggesting biofeedback was integrated smoothly into the task without hindering performance. Similarly, the surgical animation environment was rated positively, with comfort (M = 4.1 ± 0.8), support for training (M = 4.3 ± 0.6), and focus improvement (M = 4.0 ± 0.9) all scoring favourably.

4) Training Effectiveness

Participants agreed that the system improved their surgical confidence (M = 4.2 ± 0.7), accuracy (M = 4.0 ± 0.8), and conceptual understanding of surgical procedures (M = 4.1 ± 0.7). Task difficulty was considered appropriate (M = 4.0 ± 0.6), indicating a good balance between challenge and skill acquisition.

TABLE V
SUMMARY OF USER EXPERIENCE AND TRAINING METRICS

Domain	Metric	Mean ± SD	Notes
Platform Usability (PUS)	Easy to use	3.9 ± 0.8	Generally positive
	Too complex	2.1 ± 0.7	Low complexity
	Confident using	4.2 ± 0.6	High confidence
NASA-TLX	Mental demand	6.5 ± 1.2	Moderate-high
	Physical demand	5.4 ± 1.5	Moderate
	Temporal demand	4.2 ± 1.8	Moderate
	Performance	7.6 ± 1.1	High
	Effort	5.8 ± 1.3	Moderate
	Frustration	2.9 ± 1.4	Low
Biofeedback Exp.	Awareness	3.8 ± 0.9	Noticed it
	Helped focus/relax	3.6 ± 1.0	Beneficial
	Distracting	2.1 ± 0.7	Low
Animation Exp.	Comfortable	4.1 ± 0.8	Positive
	Helped training	4.3 ± 0.6	Strong support
	Improved focus	4.0 ± 0.9	Positive
Training Effect.	Confidence improved	4.2 ± 0.7	Significant gain
	Accuracy improved	4.0 ± 0.8	Positive
	Understand surgery	4.1 ± 0.7	Strong agreement
	Task difficulty	4.0 ± 0.6	Well balanced

Overall, subjective feedback indicates that the platform was well-received, with participants perceiving it as user-friendly, not overly demanding, and beneficial for both skill training and physiological awareness. These findings align with the objective improvements observed in task performance and support the feasibility of integrating HRV and EEG biofeedback into surgical simulation training.

IV. DISCUSSION

This study investigated the influence of operator relaxation and attentional state on performance in robot-user training tasks, and evaluated the extent to which biofeedback could enhance accuracy and efficiency. A custom simulation platform was developed that integrated the Phantom Omni haptic device, a Unity-based virtual environment, and real-time EEG and HRV signal acquisition. Participants completed

training and testing under three experimental conditions: no feedback (baseline), HRV biofeedback, and EEG biofeedback, with adaptive thresholds individually calibrated using mean ± standard deviation values.

Across conditions, participants demonstrated improvement from training to testing, reflected in reduced trajectory mean absolute error (MAE) and shorter task completion times. Between modalities, EEG-based feedback yielded smoother task trajectories (MAE = 15.61 mm vs. 17.48 mm for HRV) and eliminated retractions (0.0 vs. 0.6), suggesting higher sensitivity to attentional regulation. Correlation analysis reinforced these results: EEG attention values showed a moderate negative correlation with trajectory error (r = –0.34), while HRV displayed weaker but informative associations, particularly reflecting stress recovery.

Beyond performance outcomes, an alternative analysis compared HRV (RMSSD) and EEG-driven attention as candidate biofeedback signals. Bland–Altman analysis indicated that no systematic bias but high inter-individual variability, while scatter regression confirmed the absence of strong linear associations. These findings indicate that HRV and EEG are not interchangeable but capture distinct psychophysiological dimensions: EEG primarily reflects cognitive engagement, whereas HRV indexes autonomic stress regulation. This distinction underscores the potential of a multimodal biofeedback framework that integrates both signals to enhance robustness, minimize false triggers, and enable more adaptive training interventions.

A. *Evaluation*

The study successfully addressed its primary aims and objectives (see Part I), though certain limitations remain.

1) Aim 1: Influence of Stress and Attention on Accuracy

The study confirmed that stress and attention exert significant effects on accuracy in robot–user tasks. EEG-derived attentional indices demonstrated strong correlations with task accuracy, whereas HRV offered complementary but less direct insights into stress-related variability.

2) Aim 2: Efficacy of Real-Time Biofeedback

Both biofeedback modalities improved performance relative to the no-feedback baseline, with EEG feedback yielding the most consistent gains in accuracy and trajectory smoothness. HRV feedback also contributed positively, underscoring its role in capturing stress regulation, even if less immediately associated with accuracy.

3) Aim 3: User Adaptation to Task Interruptions

Participants adapted to feedback-triggered interruptions by employing strategies such as paced breathing and enhanced concentration. Nonetheless, several participants reported frustration with the animation of scalpel disappear, highlighting the importance of graded or multimodal feedback mechanisms that balance regulation with user experience.

4) Objectives

All five objectives of this study were either fully achieved or partially addressed, yielding a coherent set of findings.

First, a functional real-time surgical robot–user training platform was successfully developed, integrating Unity 3D with the Phantom Omni haptic device and incorporating EEG and HRV sensors to enable biofeedback.

Second, an adaptive biofeedback control system was implemented, with calibration thresholds (mean ± SD) individualized to each participant to determine when task interruptions (e.g., scalpel retraction) were triggered.

Third, comparative evaluation across three conditions—EEG feedback, HRV feedback, and no feedback—demonstrated that EEG-based biofeedback produced the most consistent improvements in accuracy and trajectory smoothness.

Fourth, by adopting a multimodal integration approach, the study triangulated objective performance metrics, physiological signal analysis, and subjective reports, thereby strengthening ecological validity and robustness.

Finally, statistical analyses revealed moderate but significant correlations between EEG-derived attention values and performance metrics, whereas HRV–performance associations were weaker, suggesting the need for larger samples or more sophisticated modelling approaches in future research.

The study demonstrated that real-time biofeedback, particularly EEG-based attention monitoring can enhance accuracy in surgical robot–user training tasks. While HRV contributed to performance modulation, its effects were less pronounced, suggesting that cognitive engagement is a more immediate predictor of accuracy. User adaptation strategies were evident but also highlighted the need for more refined feedback mechanisms. Future work should investigate adaptive, multilevel biofeedback frameworks to optimize both task performance and user experience.

B. *Limitations*

Despite the promising results, several limitations must be acknowledged, as they shape the interpretation of the findings and indicate directions for future research.

First, the participant cohort consisted of 24 young, healthy university students and teachers with no prior surgical background. This demographic does not reflect either surgical trainees or experienced surgeons, thereby limiting the generalisability of the results to clinical contexts. Stress responses in actual surgical procedures—where consequences are more acute—may differ substantially. In addition, several participants reported prior exposure to haptic devices, which may have facilitated faster adaptation and introduced bias into the observed performance outcomes.

Second, the fidelity of the physiological signals presented challenges. Both EEG and HRV were acquired using consumer-grade, non-clinical devices, which are inherently more vulnerable to motion artefacts and noise. For EEG, reliance on built-in noise reduction algorithms and Bluetooth/API transmission introduced processing delays that reduced the immediacy of feedback. HRV indices, calculated over short trial windows, were also less stable, constraining their reliability for continuous self-regulation.

Third, the feedback design was relatively rigid. While scalpel disappearance effectively enforced self-regulation, participants frequently perceived it as frustrating and disruptive. The absence of more graded or multimodal feedback mechanisms may have inadvertently increased cognitive load and reduced user trust in the system.

Fourth, the task design limited ecological validity. The simplified two-dimensional trajectory-tracking tasks do not capture the full complexity of surgical performance, which requires three-dimensional navigation, bi-manual coordination, and dynamic decision-making under time pressure.

Finally, the short-term scope of this preliminary study raises questions regarding the durability of the observed effects. It remains unclear whether improvements reflected genuine skill acquisition or short-lived adaptation. Without longitudinal assessment, it cannot be determined whether participants would internalise and transfer these self-regulation strategies to future training or real-world practice.

Overall, although the study demonstrates feasibility, broader applicability will require improvements in participant diversity, signal fidelity, feedback design, and long-term validation.

C. *Impact and Contributions*

Despite these limitations, the study contributes to the research field in several key ways.

This study makes several contributions to the domain of robotic surgical training and biofeedback integration. First, it demonstrates the feasibility of embedding real-time physiological monitoring into a surgical robot–user training platform with haptic interaction, illustrating how biofeedback can be employed to modulate operator states and directly impact task performance. By integrating HRV and EEG signals, the system not only enabled adaptive task interruptions but also encouraged participants to self-regulate stress and attention, thereby reinforcing psychophysiological resilience during demanding tasks.

Second, the comparative evaluation of HRV- and EEG-based feedback provides novel insights into the respective strengths and limitations of each modality. HRV proved more reliable for capturing stress-related autonomic fluctuations, whereas EEG offered finer-grained resolution of attentional engagement. Correlation analysis further revealed a moderate positive association between HRV and EEG ($r = 0.42$), indicating that the two measures capture overlapping yet distinct dimensions of operator state. This suggests that EEG should not be regarded solely as a substitute for HRV but as a complementary signal that, when combined with HRV, could underpin a more robust multimodal biofeedback framework.

Finally, by triangulating objective performance metrics, physiological signal trends, and subjective evaluations, this study underscores the broader potential of physiology-aware systems to enhance both training and real-world surgical practice. In doing so, it contributes to the emerging body of research advocating adaptive, personalized surgical training approaches that extend beyond technical skill acquisition to encompass cognitive and emotional regulation.

D. *Future Work*

Building on these limitations, several avenues for future research can refine and extend the present study.

An immediate priority is to improve the quality and responsiveness of physiological signals. Employing clinical-grade EEG and HRV devices, or implementing on-device signal processing, could mitigate noise and latency, while better synchronisation across signals would improve the reliability of feedback delivery.

The feedback mechanism also requires refinement. Rather than binary scalpel removal, future systems could adopt graded interventions—such as fading the scalpel, increasing task resistance, or providing visual or auditory cues—to encourage self-regulation without fully disrupting task performance. These approaches are likely to reduce frustration and improve user acceptance.

Beyond technical refinements, expanding the participant pool to include surgical trainees and professionals is essential for ecological validity. Comparisons between novice and expert groups could clarify how biofeedback interacts with expertise levels in realistic contexts.

Longitudinal studies are another critical step. Repeated training sessions would reveal whether participants internalise stress- and attention-regulation strategies, and whether these skills transfer to novel tasks or real-world practice.

Future work should also explore combined analyses of HRV and EEG. The present findings suggest that the two measures capture complementary aspects of psychophysiological state. Multi-signal fusion could provide a more stable composite biofeedback index, reducing false triggers and enabling more adaptive interventions.

Finally, integration of biofeedback with physical robot–user training systems represents a natural progression. Linking EEG- and HRV-driven feedback to a robotic arm would enable examination of how physiological regulation translates into real-world motor control, where factors such as instrument weight, friction, and force feedback must be addressed. This step would bridge the gap between simulation and practice, paving the way for biofeedback-assisted operating room systems that enhance surgeon performance and patient safety.

Taken together, these directions chart a path toward more realistic, adaptive, and clinically relevant biofeedback-integrated surgical training.

V. Conclusion

This study demonstrated the feasibility of incorporating EEG- and HRV-based biofeedback into a robot–user training platform, indicating that physiological regulation may enhance task accuracy, support sustained attentional focus, and reduce cognitive workload. Although limitations remain regarding participant diversity, signal fidelity, task complexity, and short-term evaluation, the findings provide evidence that biofeedback can serve not only as a monitoring instrument but also as an adaptive training mechanism. By extending the framework to more diverse populations, refining signal processing and feedback design, and ultimately integrating

with physical robot-assisted systems, this line of research establishes a foundation for next-generation, physiology-aware training platforms. Such systems have the potential to strengthen human–robot interaction, enhance surgical training, and improve patient safety.

APPENDIX

The codes for the process of bio-signals are detailed in the following GitHub link: https://github.com/Anni-Gao/BiosignalControl-Unity3DSurgicalTraningPlatform

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